

# Atharv Sandeep Patil

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## EDUCATION

University of Southern California, Los Angeles, CA

Aug 2026 – May 2028 (Expected)

Master of Science in Computer Science

Sardar Vallabhbhai National Institute of Technology, Surat, India

Aug 2020 – Jan 2024

Bachelor of Technology (Computer Science & Engineering)

CGPA: 8.75/10 (First Class with Distinction)

## TECHNICAL SKILLS

Java, Spring Boot, Python, AWS, SQL, React, JavaScript, LLM, RAG, Neo4j, Docker

## PROFESSIONAL EXPERIENCE

J.P. Morgan Chase & Co., Software Development Engineer II

Jan 2026 – Jun 2026 | Mumbai, India

- Extended a **RAG-based** system with a client-in-context feature across **5 specialized sub-agents**, enabling **~200 financial advisors** to ask client-specific questions with real-time data retrieval and improving response relevance.
- Led migration of the system from ChromaDB to **Elasticsearch** and from a Python-based RAG implementation to an agentic framework built on **Google ADK**, modernizing the architecture for improved scalability and maintainability.

J.P. Morgan Chase & Co., Software Development Engineer I

Jan 2024 – Jan 2026 | Mumbai, India

- Re-architected an offset calculator (**Spring Boot + AWS Lambda + multithreading**) to process **~600K records**, reducing runtime from ~2 hrs to ~8 min (**96% faster**) and automating report delivery via a React dashboard.
- Enhanced a **RAG-based** system to enable financial advisors query documents and databases, cutting information retrieval time by **50%+**.
- Integrated **5** legacy mainframe tables (**~15 years** of historical data) into AWS Lambda **ETL** pipelines, modernizing the data architecture to a data mesh framework and supporting mainframe decommissioning.
- Migrated the RAG service from **AWS EKS to AWS ECS**, reducing infrastructure costs by **~60%**.
- Engineered enhancements to **Spring Boot microservices** for investment account management, enabling new product launches for **~500K** private banking clients.

J.P. Morgan Chase & Co., Software Engineer Program Intern

May 2023 – Jul 2023 | Mumbai, India

- Optimized a **parser** translating internal query language into SQL using **Spring Boot**, cutting response latency and streamlining query processes.
- Added support for **Databricks SQL** dialect, reducing query errors by **~20%** and ensuring seamless compatibility.
- Constructed a nested data retrieval method in Databricks using Apache Calcite, improving retrieval **speed by ~35%**.

## RESEARCH EXPERIENCE

Undergraduate Researcher, Sardar Vallabhbhai National Institute of Technology

Jul 2023 – Dec 2023 | Surat, India

- Designed and fine-tuned an **intent detection module** to classify legal queries, enabling accurate **Cypher query generation** for retrieving case details from a **Neo4j**-based legal property graph.
- Evaluated NLU-based approach against **BERT** and **BiLSTM** baselines, achieving a significant improvement in intent detection (**F1 Score: 0.89**) while maintaining competitive entity extraction accuracy.
- Enabled **intent-based retrieval** of 9,377 Supreme Court cases using **Wit AI & Neo4j**, improving accessibility for lawyers and the public.
- Publication:** Procedia Computer Science, presented at **ICMLDE 2025**. [🔗](#)

## PROJECTS

Healthcare Platform [🔗](#), Code For Good - JPMC Hackathon

2022

- Delivered a full-stack platform used by **~200 healthcare workers** to remotely track and counsel mothers in rural India.
- Integrated a **user-friendly interface with smart logic**, aiding workers in selecting suitable medicines and vaccines, amplifying maternal and child healthcare.
- Collaborated on backend development employing **NodeJS, Express, MongoDB, and Firebase** authentication, ensuring smooth functionality and data protection.

Voyageur (Travel Website) [🔗](#), Microsoft Learn Student Ambassador

2022

- Architected a full-stack travel website using the **MERN stack**, collaborating within a team to deliver robust and scalable functionality.
- Built a custom Hotels API via web scraping, enabling streamlined hotel search and booking across 20+ Indian cities.
- Implemented **secure authentication** and **data encryption**, ensuring user access integrity and safeguarding sensitive information.

## CERTIFICATIONS & LEADERSHIP

- Amazon AWS Certified Developer – Associate (April 2026)
- Web Development Lead - Google Developer Student Clubs, NIT Surat (2022-23)
- Logistics Lead - Hack The Tank, 24-hour Pan-India Hackathon ft. Shark Tank India entrepreneurs, NIT Surat (2022-23) | 100+ participants from across India
- Video Editor Lead - Renesa, Media and Publication House, NIT Surat (2022-23)